We will cover only the following two topics in the Week 10 lecture:

1. Parametric surfaces.
2. Surface integrals of scalar fields.

Surfaces — What Are They?

Intuitively, a surface is a two-dimensional object in a higher-dimensional space.

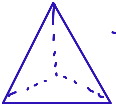
Well-known examples of a surface in $\mathbb{R}^3$ include the following:

- Planes.
- Spheres.
- Ellipsoids.
- Cylinders.
- Paraboloids.
- Hyperboloids.
- Cones. → surface with a sharp vertex

smooth surfaces

curve - 1 dimension
surface - 2 dimensions

In this course, we will study only parametric surfaces and piecewise-smooth surfaces.



tetrahedron boundary

Definition (Parametric Surfaces)

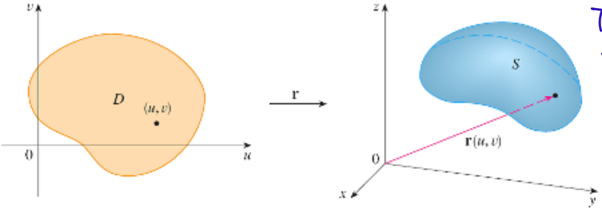
A subset S of $\mathbb{R}^3$ is called a "parametric surface" if and only if there exists an ordered pair $(\mathbf{r}, D)$ that satisfies the following three properties:

Domain or parameter space

- D is a subset of $\mathbb{R}^2$ whose interior, denoted by D°, is non-empty and connected.
- $\mathbf{r}$ is an $\mathbb{R}^3$-valued function of two real variables such that
  - $D \subseteq \text{Dom}(\mathbf{r})$,
  - $\mathbf{r}$ is continuous throughout D, and
  - $\mathbf{r}$ is one-to-one on D°.
- S is the image of D under $\mathbf{r}$, i.e., $S = \mathbf{r}[D] = \{\mathbf{r}(u, v) \in \mathbb{R}^3 \mid (u, v) \in D\}$.

$\vec{r}$: parameterization

We call such an ordered pair $(\mathbf{r}, D)$ a "surface parametrization of S".



The blue set is the image of the orange region overlap only allowed at boundary

Example

not to 4π so surface won't repeat itself

Define a function $\mathbf{r} : \mathbb{R}^2 \to \mathbb{R}^3$ by $\mathbf{r}(u, v) \overset{\text{df}}{=} (\overset{x}{2\cos(u)}, \overset{y}{v}, \overset{z}{2\sin(u)})$.

Problem: Identify and sketch the parametric surface S parametrized by $(\mathbf{r}, \overbrace{[0, 2\pi] \times \mathbb{R}}^{\text{domain}})$.

$u \in [0, 2\pi]$ $v \in \mathbb{R}$

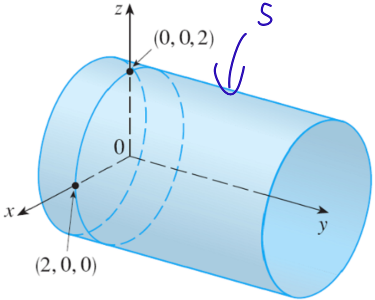
Solution

$x = 2\cos(u)$, $z = 2\sin(u)$

$\left(\frac{x}{2}\right)^2 + \left(\frac{z}{2}\right)^2 = 1$

$x^2 + z^2 = 2^2$

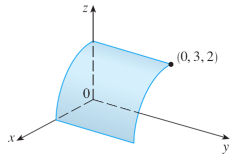
circle in xz-plane with center $(0,0)$ & radius 2

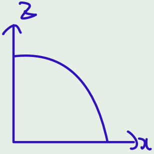


Parametric Surfaces — More Examples

Example

In Example 1, if $[0, 2\pi] \times \mathbb{R}$ were to be replaced by $\left[0, \frac{\pi}{2}\right] \times [0, 3]$, then the parametric surface obtained would be as depicted in the figure below.



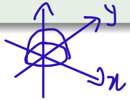


Example

Define a function $\mathbf{r} : [0, 1] \times \mathbb{R} \to \mathbb{R}^3$ by $\mathbf{r}(u, v) \overset{\text{df}}{=} \left(\overset{x}{u\cos(v)}, \overset{y}{u\sin(v)}, \overset{z}{\sqrt{1 - u^2}}\right)$.

$r\cos\theta$, $r\sin\theta$, $\sqrt{1-r^2}$

Then $(\mathbf{r}, [0, 1] \times [0, 2\pi])$ is a surface parametrization of the top hemisphere with center $(0, 0, 0)$ and radius 1.



Definition (Smooth Surface Parametrizations)

Let $(\mathbf{r}, D)$ be a surface parametrization of a parametric surface S.

Then $(\mathbf{r}, D)$ is said to be smooth if and only if it satisfies the following two properties:

1. If $\mathbf{r} = \underbrace{(X, Y, Z)}_{\text{3 components of } \vec{r}}$, then the vector functions
$$\mathbf{r}_u \stackrel{\text{df}}{=} (X_u, Y_u, Z_u) \quad \text{and} \quad \mathbf{r}_v \stackrel{\text{df}}{=} (X_v, Y_v, Z_v)$$
are <u>defined</u> and <u>continuous</u> throughout D°. interior of D
2. $\mathbf{r}_u \times \mathbf{r}_v \not\equiv (0, 0, 0)$ on D°.

$\vec{r} : D \to \mathbb{R}^3 \mid \vec{r}(u,v) = (X(u,v), Y(u,v), Z(u,v))$

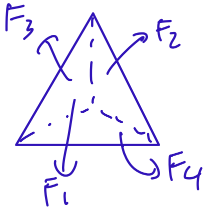
so that each point on the surface has a normal vector

Definition (Piecewise-Smooth Surfaces)

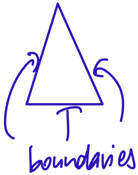
A subset S of $\mathbb{R}^3$ is called a "piecewise-smooth surface" if and only if there are finitely many parametric surfaces, $S_1, \ldots, S_k$, that satisfy the following two properties:

- $S = S_1 \cup \ldots \cup S_k$. Combine $S_1, \ldots S_k$ to form S
- We can find smooth surface parametrizations, $(\mathbf{r}_1, D_1), \ldots, (\mathbf{r}_k, D_k)$, of $S_1, \ldots, S_k$, respectively, such that $S_1, \ldots, S_k$ intersect only within can only touch at boundaries

$$\mathbf{r}_1[\text{Boundary}(D_1)], \ldots, \mathbf{r}_k[\text{Boundary}(D_k)].$$



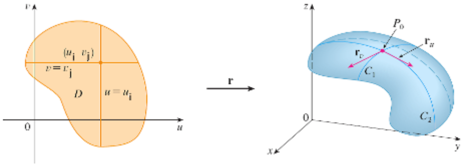
Each face looks like



Let S be a parametric surface with a smooth surface parametrization $(\mathbf{r}, D)$.

Let $(u, v) \in D^{\circ}$, and let $P_0 \overset{\text{df}}{=} \mathbf{r}(u, v)$.

Then $\mathbf{r}_{\mathrm{u}}(u, v)$ and $\mathbf{r}_{\mathrm{v}}(u, v)$ are non-~~perpendicular~~ parallel non-zero tangent vectors to S at P_0.



It follows that

$$\pm \frac{1}{\|\mathbf{r}_{\mathrm{u}}(u, v) \times \mathbf{r}_{\mathrm{v}}(u, v)\|} \cdot \underbrace{[\mathbf{r}_{\mathrm{u}}(u, v) \times \mathbf{r}_{\mathrm{v}}(u, v)]}_{\text{normal vector}}$$

are unit normal vectors to S at P_0.

$$\frac{1}{\|v\|} \cdot v = \vec{u}$$

$\downarrow$

unit vector

We have seen how to integrate a scalar field along a curve. $\int_c f\,ds$

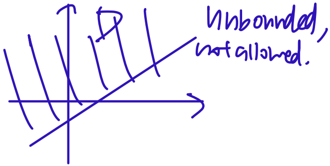
We will now see how to integrate a scalar field over a nice parametric surface.

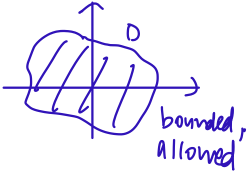
Definition (Nice Parametric Surfaces)

A parametric surface S is said to be nice if and only if it has a surface parametrization $(\mathbf{r}, D)$ that satisfies the following two properties:

- D is a bounded region in $\mathbb{R}^2$ whose boundary has zero area.
- $(\mathbf{r}, D)$ is smooth.

We call such a surface parametrization a "nice surface parametrization of S".





Definition (Surface Integral of a Scalar Field)

- Let S be a nice parametric surface.
- Let f be a continuous space scalar field such that $S \subseteq \mathrm{Dom}(f)$.

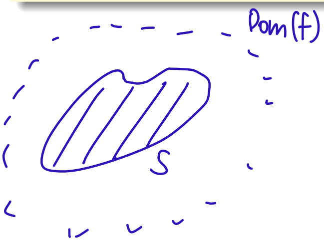
Given any nice surface parametrization $(\mathbf{r}, D)$ of S, define the surface integral of f over S, denoted by $\iint_S f \, \mathrm{d}S$, as follows:

TESTED!!

$$\iint_S f \, \mathrm{d}S \stackrel{\text{df}}{=} \iint_{D^\circ} f(\mathbf{r}(u, v)) \cdot \boxed{\|(\mathbf{r}_u \times \mathbf{r}_v)(u, v)\| \, \mathrm{d}A(u, v)}.$$

D° → in almost all cases can just write D.

Remark: This definition is independent of the choice of nice parametrization of S.



$f(\vec{r}(u,v)) = f(X(u,v), Y(u,v), Z(u,v))$

Definition (Surface Integral of a Scalar Field over a Piecewise-Smooth Surface)

- Let $S_1, \ldots, S_k$ be nice parametric surfaces.
- Suppose that $(\mathbf{r}_1, D_1), \ldots, (\mathbf{r}_k, D_k)$ are nice surface parametrizations of $S_1, \ldots, S_k$, respectively, such that $S_1, \ldots, S_k$ intersect only within

$$\mathbf{r}_1[\text{Boundary}(D_1)], \ldots, \mathbf{r}_k[\text{Boundary}(D_k)].$$

- Let S denote the piecewise-smooth surface $S_1 \cup \ldots \cup S_k$.
- Let f be a continuous space scalar field such that $S \subseteq \text{Dom}(f)$.

The surface integral of f over S, denoted by $\iint_S f \, \mathrm{d}S$, is then defined as follows:

$$\iint_S f \, \mathrm{d}S \stackrel{\text{df}}{=} \iint_{S_1} f \, \mathrm{d}S + \cdots + \iint_{S_k} f \, \mathrm{d}S.$$

these are defined by the preceding definition.

Surface Integrals — Example 1

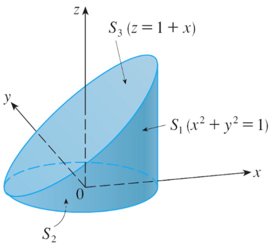
Example

Let S denote the surface boundary of the closed region in $\mathbb{R}^3$ bounded by the following three surfaces:

- The xy-plane. S_2
- The cylinder given by the equation $x^2 + y^2 = 1$. S_1
- The plane given by the equation $z = x + 1$. S_3

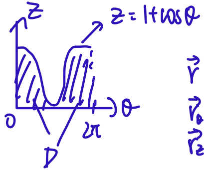
Problem: Show that S is a piecewise-smooth surface and then evaluate $\displaystyle\iint_S z \, dS$.

Solution



S_1 $\{(\cos\theta, \sin\theta, z) \in \mathbb{R}^3 \mid \theta \in [0,2\pi] \;\&\; 0 \le z \le 1+\cos(\theta)\}$

$D = \{(\theta, z) \in \mathbb{R}^2 \mid \theta \in [0,2\pi] \;\&\; 0 \le z \le 1+\cos\theta\}$



$\vec{r}(\theta, z) = (\cos\theta, \sin\theta, z)$

$\vec{r}_\theta(\theta, z) = (-\sin\theta, \cos\theta, 0)$

$\vec{r}_z(\theta, z) = (0, 0, 1)$

$$\iint_{S_1} z\,dS = \iint_D z \cdot \|\vec{r}_\theta(\theta,z) \times \vec{r}_z(\theta,z)\|\, dA(\theta,z)$$

$$\vec{r}_\theta(\theta,z) \times \vec{r}_z(\theta,z) = (\cos\theta, \sin\theta, 0)$$

$$= \iint_D z \cdot 1\, dA(\theta,z)$$

$$= \int_0^{2\pi}\int_0^{1+\cos\theta} z\, dz\, d\theta$$

$$= \int_0^{2\pi} \left[\frac{z^2}{2}\right]_{z=0}^{z=1+\cos\theta} d\theta$$

$$= \frac{1}{2}\int_0^{2\pi} [1+\cos\theta]^2\, d\theta$$

$$= \frac{1}{2}\left[\frac{3\pi}{2} + 2\sin\theta + \frac{\sin 2\theta}{4}\right]_0^{2\pi} = \frac{3}{2}\pi$$

S_2 $\vec{r}: [0,1] \times [0,2\pi] \to \mathbb{R}^3$, $\vec{r}(r,\theta) = (r\cos\theta, r\sin\theta, 0)$

xy-plane

$\vec{r}_r(r,\theta) = (\cos\theta, \sin\theta, 0)$ $\vec{r}_\theta(r,\theta) = (-r\sin\theta, r\cos\theta, 0)$

$\vec{r}_r(r,\theta) \times \vec{r}_\theta(r,\theta) = (0,0,r)$

$$\int_{S_2} z\, dS = \iint_{[0,1],[0,2\pi]} 0 \cdot \|\vec{r}_r(r,\theta) \times \vec{r}_\theta(r,\theta)\|\, dA(r,\theta)$$

$$= 0$$

(S3) $\vec{r} : [0,1] \times [0,2\pi] \to \mathbb{R}^3$, $\vec{r}(r,\theta) = (r\cos\theta, r\sin\theta, 1 + r\cos\theta)$

$\vec{r}_r(r,\theta) = (\cos\theta, \sin\theta, r\cos\theta)$

$\vec{r}_\theta(r,\theta) = (-r\sin\theta, r\cos\theta, -r\sin\theta)$

$\vec{r}_r(r,\theta) \times \vec{r}_\theta(r,\theta) = (-r, 0, r)$

$$\int_{S_3} z\, dS = \iint_{[0,1]\times[0,2\pi]} [1 + r\cos\theta] \cdot \underbrace{\|(-r,0,r)\|}_{\sqrt{2}r} \, dA(r,\theta)$$

$$= \ldots$$

$$= \sqrt{2}\pi$$

$$\int_S z\, dS = \frac{3\pi}{2} + 0 + \sqrt{2}\pi$$

$$= \left(\frac{3}{2} + \sqrt{2}\right)\pi$$

Very often, a parametric surface S is a graph surface, i.e., there exist

- a subset D of $\mathbb{R}^2$ with D° non-empty and connected, and
- a continuous real-valued function g defined on D,

such that

$$S \overset{\text{df}}{=} \mathrm{Graph}(g) = \left\{ (x, y, z) \in \mathbb{R}^3 \;\middle|\; (x, y) \in D \text{ and } z = g(x, y) \right\}.$$

In this case, parameter points (u, v) are simply written as "(x, y)".

Let D be a subset of $\mathbb{R}^2$ that satisfies the following two properties:

- D° is non-empty, connected, and bounded.
- The boundary of D has zero area.

To ensure that we can do double integration over D°

Let g be a ~~real-valued~~ cont. scalar function of two real variables with the following two properties:

- $\text{Dom}(g) = D$.
- ∇g is defined and continuous throughout D°.

∇g is bounded on D°

Let f be a continuous space scalar field such that $\boxed{S \overset{\text{df}}{=} \text{Graph}(g)} \subseteq \text{Dom}(f)$.

Then

$$\iint_S f \, dS = \iint_{D^\circ} f(x, y, \underbrace{g(x, y)}_{z}) \cdot \sqrt{1 + [g_x(x, y)]^2 + [g_y(x, y)]^2} \, dA(x, y).$$

TESTED!!

can replace this by D in most problems

Definition (Surface Area of a Nice Parametric Surface)

The surface area of a nice parametric surface S is defined as $\displaystyle\iint_S 1 \, \mathrm{d}S$.

Suppose that S is a nice parametric surface.

Let $(\mathbf{r}, D)$ be a nice parametrization of S.

TESTED!

Then the surface area of S is, explicitly,

$$\iint_{D^\circ} \|(\mathbf{r}_u \times \mathbf{r}_v)(u, v)\| \, \mathrm{d}A(u, v). \Rightarrow \text{surface area}(S)$$

Area$(D) = \iint_D 1 \, dA$

Length$(C) = \int_C 1 \, ds$

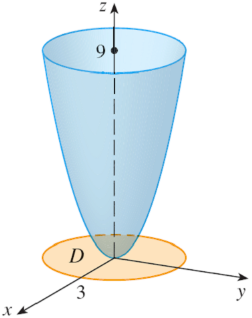
Example

Problem: Find the area of the part of the paraboloid, given by the equation

$$z = x^2 + y^2,$$

that lies under the plane given by the equation $z = 9$.

Solution



Define $g : D \to \mathbb{R}$ by $g(x,y) = x^2+y^2$

Define $\vec{r} : D \to \mathbb{R}^3$ by $\vec{r}(x,y) = (x, y, g(x,y))$

Then $(\vec{r}, D)$ is a surface parameterization of $S = \text{Graph}(g)$

Given $(x,y) \in D^\circ$, we have

$$g_x(x,y) = 2x$$

$$g_y(x,y) = 2y$$

Therefore the area of the part of the paraboloid that lies under the plane given by the equation $z=9$ is

$$\iint_S 1\,dS = \iint_{D_0} 1\sqrt{1+(2x)^2+(2y)^2}\,dA(x,y)$$ See slide 16

$$= \iint_D \sqrt{1+4(x^2+y^2)}\,dA(x,y)$$

$$= \iint_{[0,3]\times[0,2\pi]} \sqrt{1+4r^2}\cdot r\,dA(r,\theta)$$

$$= \left(\int_0^3 r\sqrt{1+4r^2}\,dr\right)\cdot\left(\int_0^{2\pi} 1\,d\theta\right)$$

$$= \left[\frac{1}{12}\sqrt{(1+4r^2)^3}\right]_0^3 \cdot \left[\theta\right]_0^{2\pi}$$

$$= \frac{37\sqrt{37}-1}{12}\cdot 2\pi$$

$$= \frac{37\sqrt{37}-1}{6}\cdot \pi$$